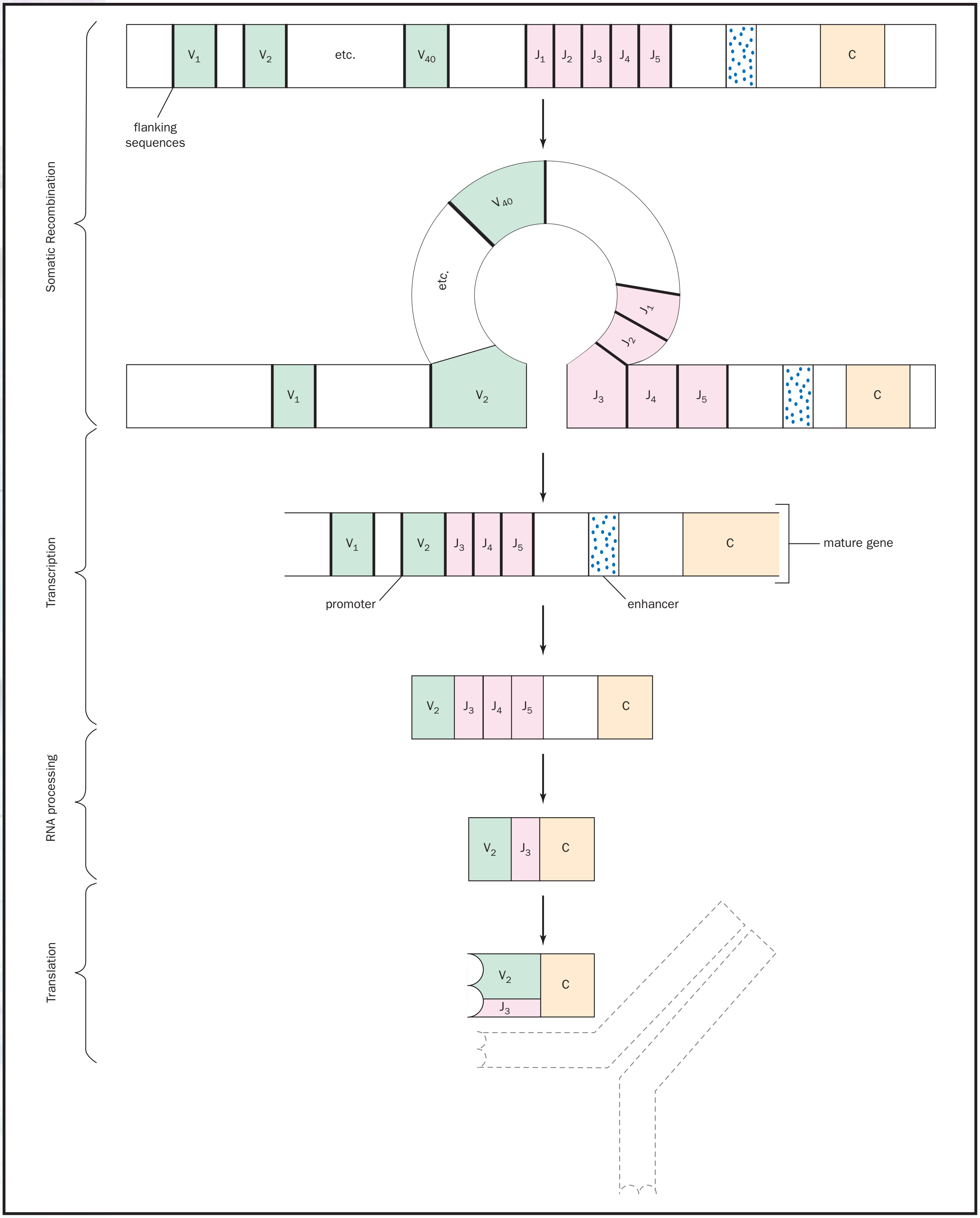


Somatic recombination in a B cell brings together a V segment and a J segment, both chosen at random. Intervening V and J segments are removed and degraded. Upstream V segments are not transcribed because their promoters are not close enough to the enhancer. Downstream J segments are transcribed but removed during RNA processing. The remaining V and J segments each code for part of the variable region of the antibody, while the C segment codes for the constant region.

There are two major classes of MHC proteins, called class I and II, with different functions in antigen presentation. Class I contains three members, each coded for by different genes, and class II contains four members. For almost every gene, there are multiple **alleles**. The number of alleles per gene ranges from a handful to more than 100. Since each person will inherit and express a unique set of MHC alleles, once again we can see the combinatorial possibilities: There are millions of different combinations of MHC alleles, and very few people are likely to have exactly the same set. This is what makes organ transplants so difficult. Matching MHC types is the key to success, but even close relatives may have different allele sets.

The MHC genes are believed to be the most allelically diverse of all human genes.

alleles particular forms of genes

Richard Robinson

Bibliography

Alberts, Bruce, et al. *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.

Janeway, Charles A., Jr., et al. *Immunobiology: The Immune System in Health and Disease*, 5th ed. New York: Garland Publishing, 2001.

Imprinting

Imprinting refers to the chemical modification of the DNA in some genes that affects how or whether those genes are expressed. One particular kind of DNA imprinting found in mammals is known as parental genomic imprinting, in which the sex of the parent from whom a gene is inherited determines how the gene is modified. While imprinting has been found in only about fifty human genes to date, some estimates suggest it may occur in several hundred more, in perhaps up to 1 percent of all genes. Imprinting defects are responsible for several human diseases, including some forms of cancer. Imprinting also occurs in other organisms, from yeast to plants to fruit flies.

Gene Expression in Imprinted and Nonimprinted Genes

Chromosomes, and the genes they contain, are inherited in pairs, with one copy of each supplied from each parent. For most genes, both members of the pair, called the maternal and paternal alleles, are used equally. Both are expressed (read by the transcription machinery to make protein) in roughly equal amounts.

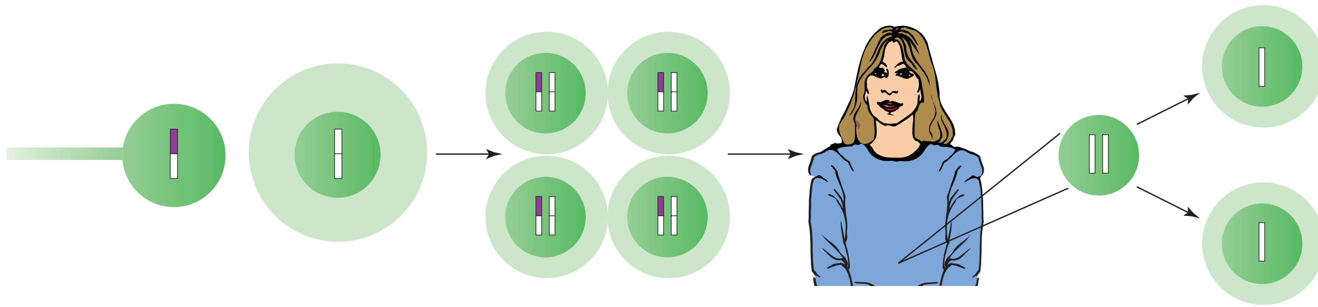
In contrast, for most imprinted genes, only one allele is expressed, while the other copy is silenced by imprinting. For some genes it is the maternal copy, for others it is the paternal copy. This is an exception to the Mendelian assumption that the two parents contribute equally to the **phenotype** controlled by autosomal genes. For some genes, both alleles are expressed, but one copy is expressed much more than the other. For some genes, the silencing occurs in some tissues but not others.

phenotype observable characteristics of an organism

Imprinted genes should not be confused with sex-linked genes, which are carried on the X or Y chromosome. Most imprinted alleles are located on **autosomes**, but are "stamped" with the sex of the parent that contributed them.

autosomes chromosomes that are not sex determining (not X or Y)

Imprinting should also not be confused with dominant and recessive alleles, in which one allele always controls the phenotype at the expense of



Imprinting silences a gene based on the sex of the parent it came from. The imprint is reset in each new generation.

recessive requiring the presence of two alleles to control the phenotype

dominant controlling the phenotype when one allele is present

the other, because of differences in the alleles themselves. The "dominance" seen in imprinting is determined by the sex of the parent contributing the allele, not any property of the allele itself. Thus, a particular allele will appear to be **recessive** when inherited from one parent, but **dominant** when inherited from the other. Such an effect, in which the expression difference is not due to the alleles but to forces acting on them from outside, is termed an "epigenetic effect."

Imprinting is thought to be responsible for many cases of incomplete penetrance, an inheritance pattern in which a dominant gene (as for a genetic disease) is not expressed in some individuals despite being present. Imprinting offers a mechanism by which a particular allele can be turned on or turned off as it is passed down through successive generations.

Timing and Mechanism of Imprinting

gametes reproductive cells, such as sperm or eggs

Although the details of imprinting are still unknown, it is clear that imprinting must occur either during the formation of the **gametes** or immediately after fertilization, while the two chromosome sets are still distinct. The imprint must be reliably passed on to each new daughter chromosome during DNA replication.

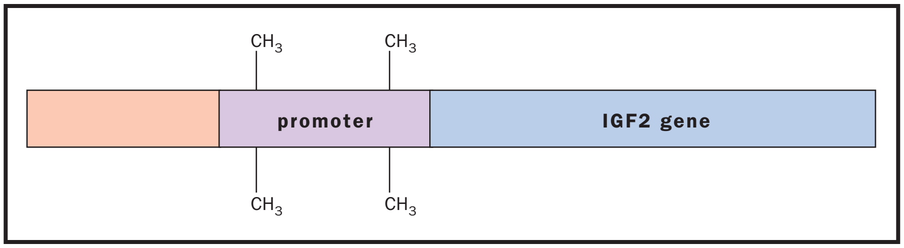
RNA polymerase enzyme complex that creates RNA from DNA template

The exact molecular mechanism of imprinting is also unknown, but it is thought to involve the modification of a gene's promoter. The promoter is the upstream region to which **RNA polymerase** binds to begin transcription. Imprinting prevents or restricts binding of RNA polymerase, thus preventing gene transcription.

nucleotides the building blocks of RNA or DNA

transcription factors proteins that increase the rate of gene transcription

One method by which a gene becomes imprinted is believed to be by the addition of methyl groups (-CH_3) to cytosine **nucleotides** in the promoter region. The evidence for methylation is strong. Methylation is a common mechanism for gene silencing, because these bulky side groups interfere with the efficient binding of the various **transcription factors** required to attract the polymerase enzyme. Methylation patterns are known to be altered during gamete formation, and are reliably passed on during replication. Further evidence comes from the observation that altered methylation patterns in some imprinted genes are associated with the aberrant expression of the normally silent allele.



Methylation of the promoter region is believed to be an important silencing region.

Example of Imprinting: The IGF2 Gene

One of the best-studied imprinted genes is the one that encodes an insulin-like growth factor called growth factor 2 (IGF2). In this gene, the paternal copy is active, whereas the maternal copy is inactive. Imagine that two parents have produced a female child. During egg formation in the mother (or shortly after fertilization), the mother's copy of the IGF2 gene is **methylated**, rendering it transcriptionally silent. The child uses only the paternal allele to make the growth factor. However, when this child makes her own eggs, neither copy of the gene will remain active, because the alleles will have been "restamped" as coming from a female. The active allele she used throughout life is passed on in an inactive form to her children.

methylated a methyl group, CH_3, added

The protein encoded by the IGF2 gene is a growth factor, which stimulates the growth of target cells. Failure to properly imprint the maternal allele, or inheritance of two copies of the male allele, can have important consequences. For example, the expression of two copies of the IGF2 gene is associated with Beckwith-Wiedemann syndrome, a growth disorder, accompanied by an increase in a type of cancer called **Wilms tumor**. Other human cancers are also associated with improper imprinting (of other genes), causing either too much or too little gene expression.

Wilms tumor a cancerous cell mass of the kidney

Uniparental Disomy and Human Disease

Inheritance of two copies of one parent's chromosome (or part of it) is called uniparental disomy, a type of chromosome aberration. Detection of uniparental disomy in individuals with genetic disorders was one of the first clues that imprinting had important developmental and medical consequences.

Prader-Willi syndrome and Angelman syndrome can both be caused by uniparental disomy of chromosome 15, which carries a maternally expressed, paternally imprinted gene. Two maternal copies of the gene causes Prader-Willi syndrome, which is marked by mild mental retardation, decreased growth of the **gonads**, and obesity. Two paternal copies of this same gene causes Angelman syndrome, marked by severe mental retardation, small head size, seizures, inappropriate laughter, and distinctive facial features. (The gene itself codes for a protein involved in degrading other proteins.) Imprinting defects can also cause these syndromes in the absence of uniparental disomy, since the result is the same: either zero or two copies of the gene are expressed.

gonads testes or ovaries

Why Imprint?

The evolutionary reason for imprinting is not yet clear, although some scientists propose that, at least in mammals, it arose from an evolutionary tug of war between males and females. In this scheme, fathers (who contribute only sperm) benefit when the embryo grows as fast as possible. Thus, silencing genes that slow down embryonic growth is in their interest, even if it depletes resources from the mother. Mothers, on the other hand, need to conserve their resources. Silencing genes that promote rapid growth is therefore in their interest. Supporting this hypothesis is the fact that many of the known imprinted genes regulate growth. Paternally expressed (maternally imprinted) genes such as IGF2 tend to promote growth, whereas maternally expressed (paternally imprinted) genes tend to inhibit it. SEE ALSO CHROMOSOMAL ABERRATIONS; FERTILIZATION; INHERITANCE PATTERNS; METHYLATION; RNA POLYMERASES.

Richard Robinson

Bibliography

Everman, David B., and Suzanne B. Cassidy. "Genomic Imprinting: Breaking the Rules." *Journal of the American Academy of Child and Adolescent Psychiatry* 39, no. 3 (March 2000): 386–389.

Greally, John M., and Matthew W. State. "Genomic Imprinting: The Indelible Mark of the Gamete." *Journal of the American Academy of Child and Adolescent Psychiatry* 39, no. 4 (April 2000): 532–535.

Paulsen, Martina, and Anne C. Ferguson-Smith. "DNA Methylation in Genomic Imprinting, Development, and Disease." *Journal of Pathology* 195, no. 1 (2001): 97–110.

Internet Resources

Geneimprint.com. <http://www.geneimprint.com/>.

Yale University School of Medicine and Yale-New Haven Hospital. <http://info.med.yale.edu>.

In Situ Hybridization

In situ hybridization is a technique used to detect specific DNA and RNA sequences in a biological sample. Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are **macromolecules** made up of different sequences of four nucleotide bases (adenine, guanine, uracil, cytosine, and thymidine). *In situ* hybridization takes advantage of the fact that each nucleotide base binds with a **complementary** nucleotide base. For instance, adenine binds with thymidine (in DNA) or uracil (in RNA) using **hydrogen bonding**. Similarly, guanine binds with cytosine.

macromolecules large molecules such as proteins, carbohydrates, and nucleic acids

complementary matching opposite, like hand and glove

hydrogen bonding weak bonding between the H of one molecule or group and a nitrogen or oxygen of another

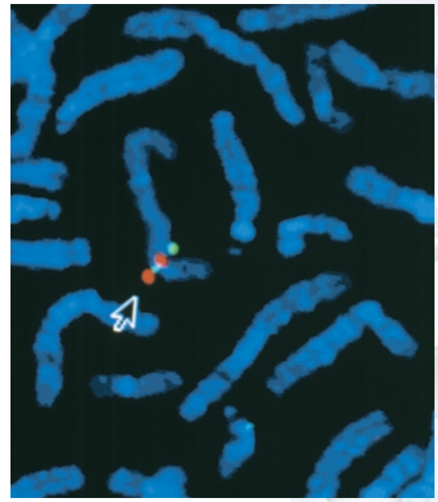
In a specialized molecular biology laboratory, researchers can make a sequence of nucleotide bases that is complementary to a target sequence that occurs naturally in a cell (in a gene, for example). When this complementary sequence is exposed to the cell, it will bind with that naturally occuring target DNA or RNA in that cell, thus forming what is known as a hybrid. The complementary sequence thus can be used as a "probe" for cellular RNA or DNA.

Thus, the term "hybridization" refers to the chemical reaction between the probe and the DNA or RNA to be detected. If hybridization is performed on actual tissue sections, cells, or isolated chromosomes in order to detect the site where the DNA or RNA is located, it is said to be done "*in situ*." By contrast, "in vitro" hybridization takes place in a test tube or other apparatus, and is used to isolate DNA or RNA, or to determine sequence similarity of two nucleotide segments.

Application of the Probe for DNA or RNA to Tissues or Cells

In situ hybridization allows us to learn more about the geographical location of, for example, the messenger RNA (mRNA) in a cell or tissue. It can also tell us where a gene is located on a chromosome. Obviously, a detection system must be built into the technique to allow the **cytochemist** to visualize and map the geography of these molecules in the cells in question.

When *in situ* hybridization was first introduced, it was applied to isolated cell nuclei to detect specific DNA sequences. Early users applied the techniques to isolated chromosomal preparations in order to map the location of genes in those chromosomes. The technique has also been used to detect viral DNA in an infected cell. *In situ* hybridization of RNA has also been used to show that RNA synthesis (transcription) occurs in the nucleus, while protein synthesis (translation) occurs in the **cytoplasm**.



The arrow points to a chromosome section illuminated by the FISH procedure.

cytochemist chemist specializing in cellular chemistry

cytoplasm the material in a cell, excluding the nucleus

protocol laboratory procedure

Conditions that Promote Optimal *In Situ* Hybridization

Hybrid probes are known as cDNA or cRNA, because they are complementary to the target molecule. In developing an *in situ* hybridization **protocol**, it is vital to learn optimal temperatures and times needed for formation of the hybrid between the cDNA probe or cRNA probe and unique RNA or DNA in the cell. The optimal hybridization temperature depends on several factors, including the types of bases in the target sequence and the concentration of certain ingredients in the media. The concentration of cytosine and guanine in the sequence plays an important role. A cytochemist will use these factors to calculate optimal temperature when planning the experiment. One must be as careful in setting up an *in situ* hybridization experiment as one is when setting up a test tube hybridization assay. The cytochemist and molecular biologist work together to optimize the conditions.

Another vital consideration in developing good *in situ* hybridization techniques is the specificity of the probe itself. If the investigators know the exact nucleotide sequence of the mRNA or DNA in the cell, they can design a complementary probe and have it made in a molecular biology lab. However, if the investigators do not know the exact sequence, they may try a sequence that is as close to exact as possible (such as from a related species). For example, they might try a cDNA probe for a mouse DNA sequence on a tissue preparation from a rat. This may or may not work because, if over 5 percent of the base pairs are not complementary, the probe will bind only loosely to the target. This loose binding may cause it to be dislodged in the washing or detection steps and hence the reaction will not be detected, or only some of the sites may be detected and the labeling will not accurately reflect all of the target sites.

Application of Tools to Detect the Hybridization Sites

Forming the hybrid is not sufficient to allow it to be mapped, because the molecules are too small to be seen in the microscope. Thus, to map the geographical distribution of the gene or gene product, the cytochemist applies sensitive detection systems that allow the hybrid to be seen. This is done by labeling the cDNA or cRNA probe itself with a molecule that can either be visualized directly (such as fluorescein, which glows when exposed to fluorescent light in a microscope) or indirectly (such as radioactive sulfur, which can be detected by autoradiography; or biotin, which can be detected by avidin or by specific **antibodies** to biotin). Whichever type of molecule is chosen, it should be small, so that it does not interfere with the hybridization process.

antibodies immune system proteins that bind to foreign molecules

The molecule used to visualize the hybrid is called the reporter molecule because it "reports" the site of the hybridization of the probe to the cellular DNA or RNA. Many different detection systems are available to cytochemists, employing a wide variety of reporter molecules. These include fluorescent compounds, colloidal gold compounds, or enzyme reactions or radioactive elements. Cytochemists will choose the most sensitive detection system that is also appropriate for their laboratories. For example, laboratories that do not want to work with radioactive compounds may choose one of the many nonradioactive methods.

One of the best-known nonradioactive *in situ* hybridization methods is FISH or "fluorescence *in situ* hybridization." It allows the detection of many genes in a chromosome or a nucleus, and different combinations of fluorescent reporter molecules are used to produce different colors. Using FISH, a veritable rainbow of colors can be used to map the location of genes on chromosomes. Another method uses enzyme reactions to form a product over the site of the mRNA or DNA in the cell. One can use different enzymes or enzyme substrates in the detection system and thus detect multiple gene products in the same tissue or cells.

Preserving the Tissues or Cells and Preventing Loss of DNA or RNA

Another important consideration in developing *in situ* hybridization technology involves the preservation of the cells or chromosomes. It is important to preserve the morphology (shape) and geographical site in the cell or chromosome where the target DNA or mRNA is located. Investigators may choose to use frozen sections, or they may treat cells or tissues with fixatives that cross-link proteins and stabilize cell structure. This prevents destruction during the hybridization and washing protocols.

Preserving DNA is easy because it is a highly stable molecule. However, preserving RNA is much more difficult because of a very stable enzyme called RNase, which may be found on glassware, in lab solutions, or on the hands of the cytochemist. RNase will quickly destroy any RNA in the cell or the RNA probe itself. Thus, investigators that work with RNA must use sterile techniques, gloves, and solutions to prevent RNase from contaminating and destroying the probe or tissue RNA.

Controlling the Specificity of the Cytochemical Assays

Good cytochemists know that experimental results must be checked and verified. For this reason, in addition to testing for reactions with their target DNA or RNA, they also test for reactions with unrelated nucleotide sequences. Likewise, they test for reactions with other components of the detection systems.

There are a series of controls that are run that detect if the labeling pattern is due to the proper sequence of reactants. For example, if the cytochemist leaves out the probe in the hybridization solution, there should be no reaction. Similarly, if the cytochemist changes the sequence of the probe, or uses a noncomplementary probe, there should be no reaction (unless that new sequences reacts with another sequence in the cell). Tests of the detection system must also be run by leaving out one or more components to learn if the reaction is dependent totally on the complete sequence of reactants. SEE ALSO DNA; NUCLEOTIDE; RNA.

Gwen V. Childs

Bibliography

Bloom, Mark V., Greg A. Freyer, and David A. Micklos. *Laboratory DNA Science: An Introduction to Recombinant DNA Techniques and Methods of Genome Analysis.* Menlo Park, CA: Addison-Wesley, 1996.

Brahic, M., and A.T. Haase. "Detection of Viral Sequences of Low Reiteration Frequency by *In situ* Hybridization." *Proceedings of the National Academy of Science USA* 75 (1978): 6125–6127.

Buongiorno-Nardelli, S., and F. Amaldi. "Autoradiographic Detection of Molecular Hybrids between rRNA and DNA in Tissue Sections." *Nature* 225 (1970): 946–948.

Childs, G. V. "*In situ* Hybridization with Nonradioactive Probes." In *Methods in Molecular Biology*, vol. 123: *In situ* Hybridization Protocols, I. A. Darby, ed. Totowa, NJ: Humana Press, Inc, 1999.

Gall, J. G., and M. Pardue. "Formation and Detection of RNA-DNA Hybrid Molecules in Cytological Preparation." *Proceedings of the National Academy of Science USA* 63 (1969): 378–383.

Haase, A.T., P. Venture, C. Gibbs, and W. Touretellotte. "Measles Virus Nucleotide Sequences: Detection by Hybridization *In situ.*" *Science* 212 (1981): 672–673.

John, H. A., M. L. Birnstiel, and K. W. Jones. "RNA-DNA Hybrids at the Cytological Level." *Nature* 223 (1969): 582–587.

Inbreeding

Inbreeding is defined as mating between related individuals. It is also called consanguinity, meaning "mixing of the blood." Although some plants successfully self-fertilize (the most extreme case of inbreeding), biological mechanisms are in place in many organisms, from fungi to humans, to encourage cross-fertilization. In human populations, customs and laws in many countries have been developed to prevent marriages between closely related individuals (e.g., siblings and first cousins). Despite these proscriptions, genetic counselors are frequently presented with the question "If I marry my cousin, what is the chance that we will have a baby who has a disease?" The answer is that when two partners are related their chance to have a baby with a disease or birth defect is higher than the background risk in the general population.

Increased Disease Risk

Many genetic diseases are recessive, meaning only people who inherit two disease **alleles** develop the disease. All of us carry several single alleles for genetic diseases. Since close relatives have more genes in common than unrelated individuals, there is an increased chance that parents who are closely related will have the same disease alleles and thus have a child who is homozygous for a recessive disease.

alleles particular forms of genes

locus site on a chromosome (plural, loci)

deleterious harmful

For instance, cousins share approximately one-eighth or 12.5 percent of their alleles. So, at any **locus** the chance that cousins share an allele inherited from a common parent is one-eighth. The chance that their offspring will inherit this allele from both parents, if each parent has one copy of the allele, is one-fourth. Thus, the risk the offspring will inherit two copies of the same allele is $1/8 \times 1/4$, or 1/32, about 3 percent. If this allele is **deleterious**, then the homozygous child will be affected by the disease. Overall, the risk associated with having a child affected with a recessive disease as a result of a first cousin mating is approximately 3 percent, in addition to the background risk of 3 to 4 percent that all couples face.

Inbreeding can be measured by the inbreeding coefficient (often denoted F). This is the probability that two genes at any locus in one individual are identical by descent (have been inherited from a common ancestor). F is larger the more closely related the parents are. For example, the coefficient of inbreeding for an offspring of two siblings is one-fourth (0.25), for an offspring of two half-siblings it is one-eighth (0.125), and for an offspring of two first cousins it is one-sixteenth (0.0625). (This is a different calculation than the calculation of shared alleles between cousins, above.)

In general, inbreeding in human populations is rare. The average inbreeding coefficient is 0.03 for the Dunker population in Pennsylvania and 0.04 for islanders on Tristan da Cunha. Inbreeding occurs in both those populations. Some isolated populations actively avoid inbreeding and have maintained low average inbreeding coefficients even though they are small. For example, polar Eskimos have an average inbreeding coefficient that is less than 0.003.

Beneficial changes can also come from inbreeding, and inbreeding is practiced routinely in animal breeding to enhance specific characteristics, such as milk production or low fat-to-muscle ratios in cows. However, there can often be deleterious effects of such selective breeding when genes controlling unselected traits are influenced too. Generations of inbreeding decrease genetic diversity, and this can be problematic for a species. Some endangered species, which have had their mating groups reduced to very small numbers, are losing important diversity as a result of inbreeding.

Genetic Studies of Inbred Populations

Inbred populations can offer a rich resource for genetic studies. They have the advantage of often being relatively homogeneous in both their genetics and environment. A method that has been used successfully to identify several recessive mutations in inbred groups is homozygosity mapping.

This approach looks for regions of alleles at genetic loci that are linked to one another and are homozygous. With inbreeding, there is an increased chance that, in an affected individual, the two alleles at the disease locus will have descended from a common ancestor. Therefore tightly linked markers (identifiable DNA segments) surrounding the disease locus will also tend to come from the same ancestral chromosome and thus be identical on both homologous chromosomes.

Together with colleagues, Erik Puffenberger, a research scientist and laboratory director at the Clinic for Special Children in Strasburg, Pennsylvania, capitalized on the inbreeding in a large Mennonite kindred to identify the location of a gene for Hirschprung disease on chromosome 13. In this family, parents of an affected child are, on average, related as closely as second or third cousins. The region was located because, true to theory, affected individuals shared alleles that were identical by descent at the region containing the disease gene. SEE ALSO Founder Effect; Inheritance Patterns; Pedigree; Population Genetics.

Eden R. Martin and Marcy C. Speer

Bibliography

Cavalli-Sforza, Luigi L., and Walter F. Bodmer. *The Genetics of Human Populations.* Mineola, NY: Dover Publications, 1999.

Puffenberger Erik G., et al. "Identity-by-Descent and Association Mapping of a Recessive Gene for Hirschprung Disease on Human Chromosome 13q22." *Human Molecular Genetics* 3 (1994): 1217–1225.

Individual Genetic Variation

"Variety is the spice of life," or so the saying goes. In fact, it is probably more precise to say that variety is the key to life. It is genetic variation that contributes to the diversity in **phenotype** that provides for richness in human variation, and it is genetic variation that gives evolution the tool that it needs for selection and for trying out different combinations of alleles and genes.

phenotype observable characteristics of an organism

locus site on a chromosome (plural, loci)

polymorphic occurring in several forms

deleterious harmful

Some variation is directly observable. Some examples of human genetic variants include the widow's-peak hairline, which is dominant to non-widow's peak; free earlobe, which is dominant to attached earlobe; facial dimples, which are dominant to no facial dimples; and tongue-rolling, which is dominant to non-tongue-rolling. Another example is the ability to taste phenylthio-carbamide (PTC), a bitter-tasting substance. Seven out of ten people can taste the bitterness in PTC paper, and the ability to taste it is dominant to nontasting. Much variation at the genetic level, however, is not observable just by looking at someone. Even these "invisible" traits can nonetheless be scored in the laboratory.

This woman has the inherited trait that allows her to roll her tongue.

Variation and Alleles

Any **locus** having two or more alleles (variant forms of particular genes), each with a frequency of at least 1 percent in the general population, is considered to be **polymorphic**. The difference between two alleles may be as subtle as a single base-pair change, such as the thymine-to-alanine substitution that alters the B chain of hemoglobin A from its wild type to its hemoglobin sickle cell state. These single nucleotide polymorphisms are termed "SNPs" ("snips").

As of March 2002, in the approximately 3.2 billion base pairs of DNA, approximately 3.2 million SNPs have been identified. Some base-pair changes have no **deleterious** effect on the function of the gene; nevertheless, these functionally neutral changes in the DNA still represent different alleles. Alternatively, allelic differences can be as extensive as large, multi-codon deletions, such as those observed in Duchenne muscular dystrophy.

Genetic variations cause differences in appearances such as dimples, widow's peaks, and even hairy ears.

Ear wax consistency is due to single gene with two alleles (wet vs. dry), located near the chromosome 16 centromere.

germ cells cells creating eggs or sperm

somatic nonreproductive; not an egg or sperm

Genetic variation is transmissible from parent to child through **germ cells**, and it can occur anew via mutation in either germ cells or in **somatic** cells. Interestingly, new mutations occur twice as frequently in sperm as in eggs, probably because so many more cell divisions are required to make sperm than eggs.

Scoring Variation in the Lab

Differences in alleles can be scored via laboratory testing. The ability to score allele differences accurately within families, between families, and between laboratories is critically important for linkage analysis in both simple Mendelian and genetically complex common disorders. Linkage analysis traces coinheritance of a disease gene and polymorphic markers such as SNPs to discover where in the chromosomes the disease gene is located.

Allele scoring strategies may be as simple as noting the presence (+) or absence (-) of a deletion or point mutation, or as complicated as assessing the allele size in base pairs of DNA. The latter application is common when highly polymorphic; microsatellite repeat markers are used in linkage analysis.

When genetic counselors talk to patients and families about mutations (a type of genetic variation) that are present in themselves or their children, they are careful to point out that each individual is estimated to carry between five to seven deleterious alleles that, in the right combination with other genes or with specific environmental influences, can lead to disease. Most of us do not know which deleterious genes we carry. Some are recessive and do not influence the genotype unless paired with a second recessive allele. Thus, these alleles will not be noticed without genetic analysis. And, genetic counselors are careful to avoid the term "mutation," because it is potentially stigmatizing. When speaking with patients, they prefer to use the more neutral term "variant." SEE ALSO Disease, Genetics of; Genotype and Phenotype.

Marcy C. Speer

Bibliography

Internet Resources

SNP Consortium, Ltd. <http://brie2.cshl.org>.

"Human Genome Project Information." U.S. Department of Energy. <http://www.ornl.gov/hgmis/publicat/primer/intro.html>.

Information Systems Manager

An information systems manager (ISM) is a professional whose skills are needed to handle the large amounts of information generated by and analyzed in the modern genetics laboratory. A successful information systems manager needs to be experienced with the technical aspects of computer hardware and networking systems. The daily work may involve managing a team of information technology workers, so leadership and management skills are usually a necessary qualification. Like most jobs in the scientific and technical areas, this one involves interacting regularly with staff, other units, upper management, and scientists, so that proficiency in oral and written communication is essential. Although an information systems manager is not usually required to have a deep understanding of the genetic sciences, the ISM must be well acquainted with the information needs of the organization and must possess advanced technical knowledge.

The training needed for a career as an information systems manager begins with formal education and training in computer programming, database fundamentals, systems analysis and design, data communication, and networking, all at the undergraduate level. A major, or at least a concentration, in information systems management at either the bachelor's or master's degree level will also provide a solid foundation for a position as an ISM. Many information systems managers begin their career as systems analysts, programmers, or computer engineers, or were in other computer-related positions. An in-depth understanding of computer technology and applications will make the candidate attractive for many other careers as well, since the projected outlook for all computer-related professions is continued growth through 2008.

The duties of an ISM can be quite diverse. They include setting up networks, including the installation of lines, hardware, and software; administering servers; programming; and setting up intra- and internets. ISMs may even be called upon to do some Web page design. An information systems manager is not an entry-level position, as the ISM has considerable administrative duties. The job requires hiring, training, assigning, and supervising computer specialists so that each employee does his or her particular duty toward meeting the organization's goals. In addition, ISMs need strong budgeting skills and the ability to communicate with contractors, suppliers, and financial groups, for they are involved in making decisions concerning equipment purchases.

The majority of ISMs do most of their work from an office, but may be called upon to go to the client's site to set up networks, hardware, or software. This may include doing their work in research laboratories, where they may be exposed to hazards arising from the investigations being conducted there. ISMs often work under stringent time and budgetary contraints. It is not unusual for ISMs to work more than forty hours per week.

The job can be rewarding when the organization makes progress in meeting their goals. There is usually plenty of room for personal growth, as many ISMs are advanced to higher managerial positions. ISMs typically enjoy a good salary and a substantial benefits package. Salaries usually range from $45,000 to $120,000 per year, with most earning between $50,000 and $95,000. The amount earned depends on education, the amount of experience, and whether the job is in the public, academic, or private sector. Workers in state government usually earn the least, with a median annual income of $63,500, while workers in industrial settings earn the most, with a median annual income of $87,500. SEE ALSO BIOINFORMATICS; COMPUTATIONAL BIOLOGIST; STATISTICAL GENETICIST.

Judith E. Stenger

Bibliography

Ahituv, Niv, and Seev Neumann. *Principles of Information Systems for Management.* Dubuque, IA: Wm. C. Brown Company, 1983.

Alter, Steven. *Information Systems: A Management Perspective*, 2nd ed. Menlo Park, CA: Benjamin/Cummings Publishing Company, 1996.

Internet Resource

U.S. Bureau of Labor Statistics. *Occupational Outlook Handbook, 2000–2001 edition.* U.S. Department of Labor. Washington, DC, 2000. <http://stats.bls.gov/ocohome.htm>.

Inheritance, Extranuclear

Less than a decade after the rediscovery of Mendel's laws describing the inheritance of genes in the nucleus, hereditary traits were discovered that obey a different set of laws. The genes involved in this non-Mendelian pattern of inheritance reside outside the nucleus, in the cytoplasm of the cell. Specifically, they were found to reside in **mitochondria**, **chloroplasts**, or intracellular symbiotic bacteria. Those genes play important roles in the cell. Mutations in extranuclear genes are responsible for some hereditary diseases in humans and other organisms, are used in plant breeding, and are used to study population genetics and evolution.

mitochondria energy-producing cell organelle

chloroplasts the photosynthetic organelles of plants and algae

cytoplasm the material in a cell, excluding the nucleus

Genes in Mitochondria and Chloroplasts

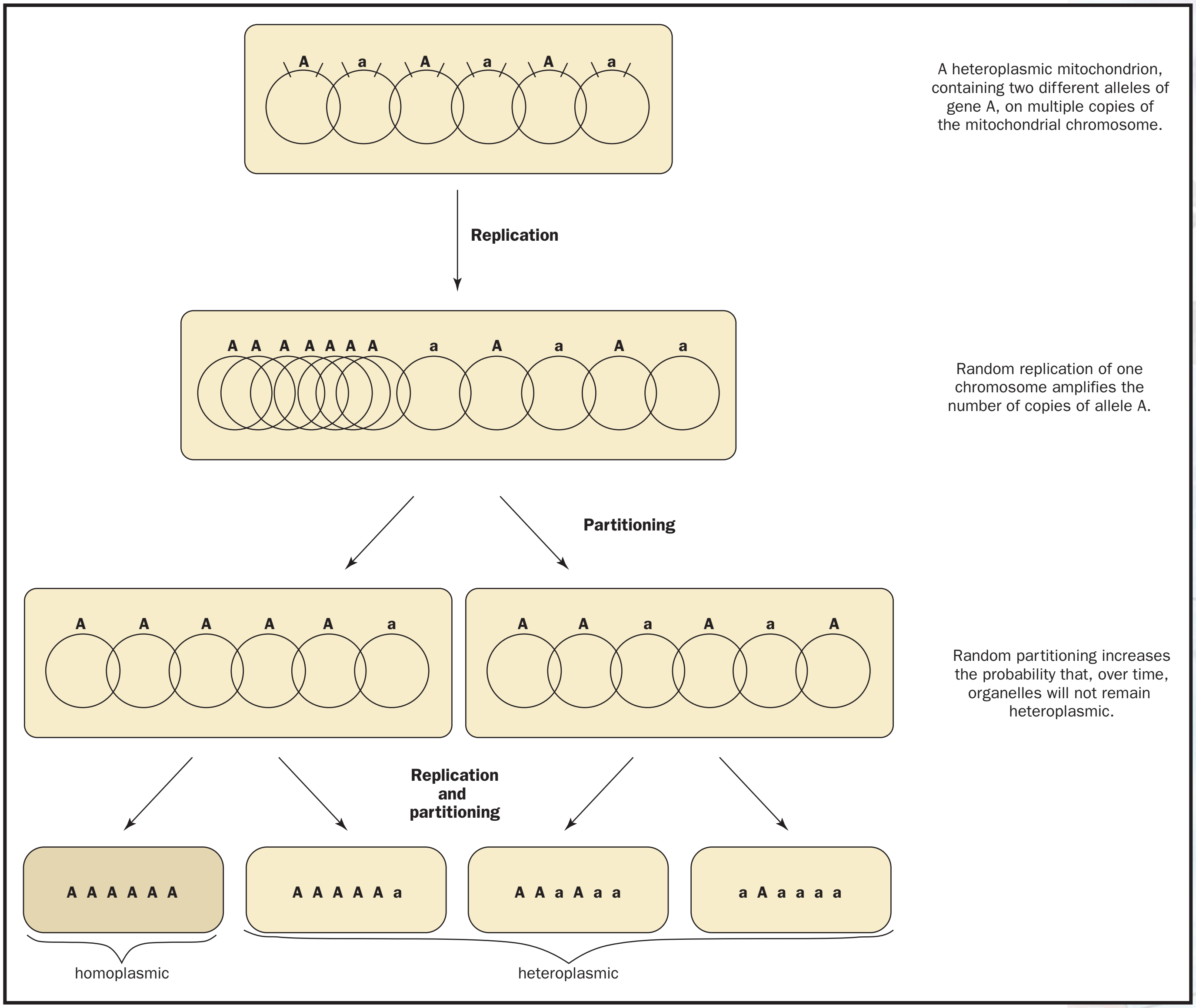
The **cytoplasm** of most eukaryotic cells contain organelles called mitochondria, where energy is extracted from food molecules and stored in ATP (adenosine triphosphate) for later use throughout the cell. Virtually all of the oxygen we use is consumed by our mitochondria.

Mitochondria contain their own DNA molecules (mitochondrial DNA, or mtDNA). These molecules carry a few dozen genes that are essential for energy metabolism. For example, the *cob* gene carries the instructions for making a protein, cytochrome *b*, which is an important component of the electron transport system in mitochondria. All the other proteins and RNAs encoded by mtDNA genes are also used in energy metabolism. However, many other key proteins for energy metabolism are encoded by nuclear genes. These are synthesized elsewhere in the cell and imported into the mitochondria. In fact, while the mtDNA genes are absolutely essential for the aerobic production of energy, the majority of all mitochondrial components derive from nuclear genes.

In addition to mitochondria, the cells of plants and algae also contain organelles called chloroplasts, in which photosynthesis takes place. Like the mitochondria, chloroplasts contain DNA molecules (chloroplast DNA, or cpDNA). The cpDNA molecules have genes that encode some of the proteins needed for photosynthesis. Also like the mitochondria, the majority of components needed for photosynthesis are made outside the chloroplast, using information from nuclear genes.

Endosymbiotic Origin of Mitochondria and Chloroplasts

Mitochondria and chloroplasts are self-replicating organelles: They can arise only by division of preexisting mitochondria or chloroplasts. DNA molecules are replicated and divided up among the daughter organelles after division. As a result, organelle genes show hereditary continuity from cell to cell and from parent to offspring, as do the genes in the nucleus. In



fact, the ancestry of mtDNA and cpDNA can be traced back to intracellular **symbionts** (termed "endosymbionts").

Early in evolutionary history, an ancestor of the eukaryotes ingested an ancestor of the aerobic α-proteobacteria. This bacterium avoided digestion and became a permanent resident of the host cell, dividing within it and providing it with energy from **aerobic** metabolism. Gradually, over millions of years, the endosymbionts transferred most of their genes to the host nucleus, becoming completely dependent on the host cells. The host cells, in turn, came to depend on the symbiont for aerobic energy production.

Much later, an ancestor of the modern green algae and plants ingested a cyanobacterium capable of photosynthesis. Gradually this endosymbiont also lost most of its genes to the nucleus and became dependent on the host cell, while providing the host with energy from photosynthesis. The resulting **organelles** are self-replicating, like the original symbiont.

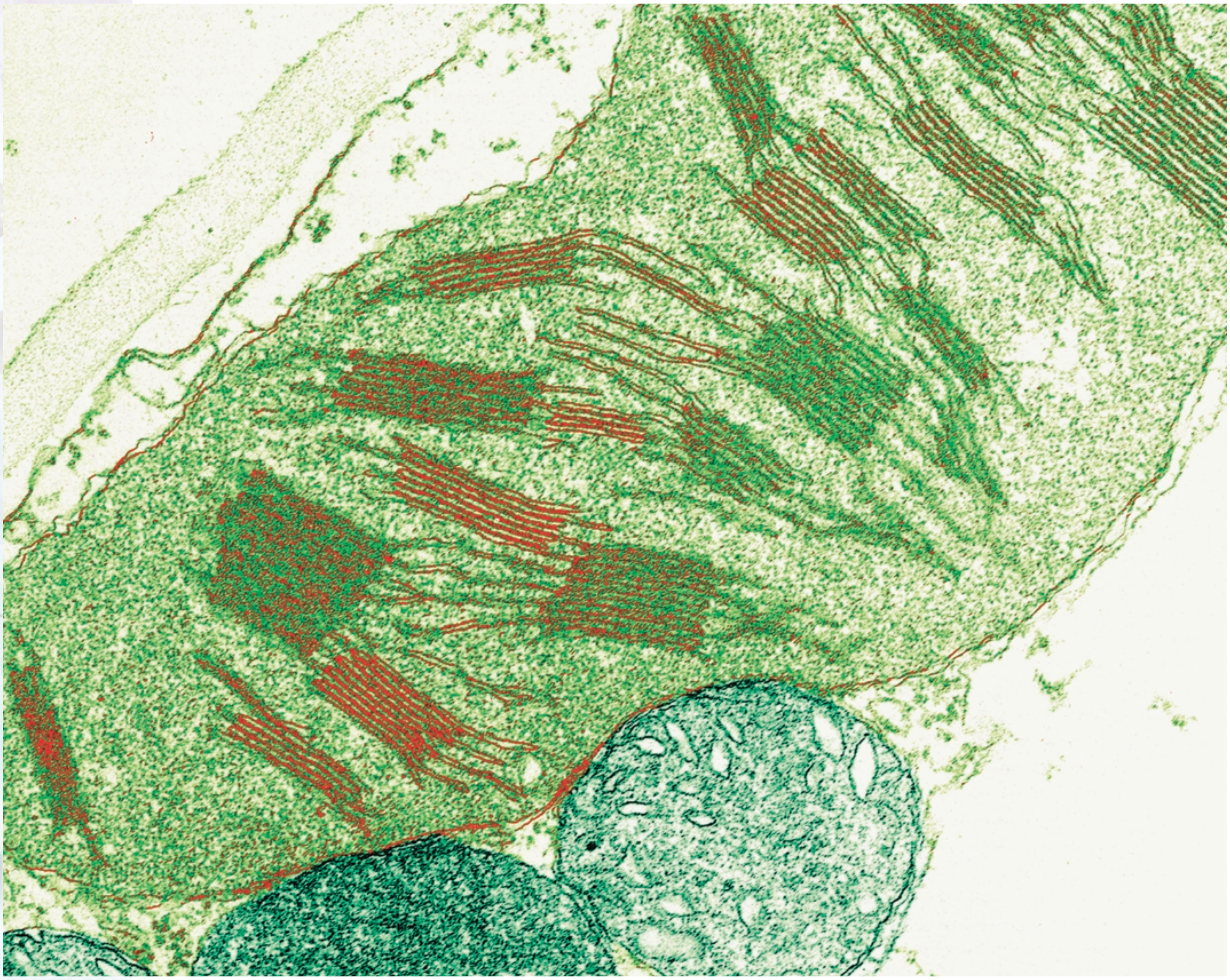
Because mitochondria and chloroplasts originated as endosymbiotic bacteria, their genomes differ from the nuclear genome in several important

Multiple copies of the mitochondrial genome exist in each organelle, which may bear different alleles. Replication and partitioning can create mitochondria with only one allele type.

symbionts organisms living in close association with other organisms

aerobic with oxygen, or requiring it

organelles membrane-bound cell compartments



Tracked mutations in extranuclear genes, as in the chloroplasts of the plant cell shown here can be used to study the evolution of a species.

ways. First, all of the organelle genes are located on a single, circular DNA molecule. Second, the genes are virtually contiguous, with little or no intergenic DNA. Third, the gene coding sequences are continuous. In other words, there are no (noncoding) introns separating gene-coding sequences. Also, each organelle has many copies of the DNA molecule, and each cell usually has more than one organelle.

Plant cells commonly have from two to several hundred chloroplasts, while animal cells often have hundreds of mitochondria. As a result, each cell has hundreds or thousands of mtDNA or cpDNA molecules, and hence of each mitochondrial or chloroplast gene. In effect, each cell contains a small population of organelle genes. This is in contrast to the nucleus where, with few exceptions, there are only two copies of each chromosome and gene (or one copy in haploid cells).

Non-Mendelian Inheritance of Organelle Genes

The inheritance of genes in chloroplasts and mitochondria differs from that of nuclear genes in several ways. For instance, organelle genes are characterized by uniparental inheritance. During sexual reproduction, the nuclear genes are inherited from both parents (biparental inheritance). In contrast, the organelle genes are often inherited from only one parent. In animals, this is usually the female parent (maternal inheritance). Mitochondrial genes are inherited maternally because the egg is much larger than the sperm and contains tens of thousands of mtDNA molecules, while the sperm contains only a hundred or so mtDNA molecules. As a result, paternal genes are greatly outnumbered by the maternal genes and can be lost during random replication or other chance events. In addition, in some animals the mitochondrial or mtDNA molecules in the sperm are singled out for degradation in the egg.

Organelle genes are inherited maternally in most plants. In some conifers, mitochondrial genes are inherited maternally, whereas chloroplast genes are inherited paternally. In other conifers, both organelle genomes are inherited paternally. Some other plants (for example, the geranium) and some fungi and protists show a mixed pattern of inheritance. In these organisms, some offspring from a mating inherit organelle genes from only one parent, some only from the other, and some from both parents.

Another way in which extranuclear inheritance differs from nuclear genes is that organelles show vegetative segregation. During the mitotic divisions that produce an adult **eukaryote** from a single cell, each daughter cell receives one copy of each preexisting nuclear chromosome and gene. The result is that if a cell is heterozygous (has two different versions or alleles of a gene in the nucleus), all of the daughter cells are also heterozygous.

eukaryote organism with cells possessing a nucleus

In contrast, different alleles of organelle genes segregate from each other during mitosis. For example, a plant egg with a mixture of normal green and mutant white chloroplasts develops into a plant with a mixture of cells, some of which will contain all green chloroplasts, whereas others will contain all white chloroplasts. This process is called vegetative segregation because it was first discovered in plants, where green and mutant white chloroplasts were observed to segregate during vegetative growth. However, it is now known to occur in all eukaryotes.

Vegetative segregation is the result of two remarkable features of organelle genes. One is random replication. Recall that an organelle contains many copies of its DNA molecule. When mtDNA or cpDNA molecules are replicated before cell division, individual molecules are randomly selected for replication until the total number of molecules has doubled. Consequently, some molecules, and some alleles, are replicated more than others.

The other feature is random partitioning. When an organelle divides, the mtDNA or cpDNA molecules are partitioned (divided up) randomly between the daughter organelles. The result is that heteroplasmic mitochondria or chloroplasts produce homoplasmic daughter organelles with a certain probability in each generation. (An organelle is said to be heteroplasmic if it contains two or more forms of a particular gene. If all the gene copies are identical, it is homoplasmic.) Moreover, when a cell divides, the organelles are partitioned randomly between the two daughter cells, so that a heteroplasmic cell can produce homoplasmic daughters. Over a large number of cell divisions, random replication and random partitioning result in the complete replacement of heteroplasmic cells by homoplasmic cells.

Plasmids

Bacteria often carry small circular DNA molecules called plasmids. These molecules sometimes carry genes that have important properties; for example, some plasmid genes make the host cell resistant to antibiotics, with important consequences for human health. Plasmids are widely used in genetics labs in the process of cloning genes. Like organelle DNA in eukaryotes, plasmid molecules are replicated independently of the cell chromosome, and one cell can contain many copies. Plasmid DNA molecules are replicated randomly and partitioned randomly to daughter cells, just like organelle genes. As a result, a cell with two different **genotypes** of plasmids may produce daughter cells with only one or the other. This process, analogous to vegetative segregation, is called plasmid incompatibility.

genotype set of genes present

Genes in Intracellular Symbionts

Many eukaryotes harbor intracellular symbiotic bacteria as well as organelles. These are usually inherited from only one parent and may have significant effects on their host. Many insects have endosymbionts that are inherited only through the female **germ line**. Well-studied examples are bacteria of the genus *Buchnera* in aphids. Since they took up residence in insect cells, *Buchnera* have lost a number of genes, just like the early ancestors of mitochondria and chloroplasts. Protists also harbor hereditary symbiotic bacteria. An example is a bacterium called *kappa*, found in *Paramecium*. *Kappa* makes a toxin that is secreted by its host and kills other *Paramecium* cells that do not contain *kappa*.

germ line cells giving rise to eggs or sperm

The Practical Importance of Extranuclear Genes

Mutations of mitochondrial genes have been found to cause a number of hereditary diseases in humans. Many of these mutations lead to defects in muscles, including the heart, and the nervous system. Because mitochondrial genes are found in nearly all eukaryotes, they are often used to trace the evolutionary history of organisms, including humans. Chloroplast genes are used for evolutionary studies in plants and algae. When organelle genes are inherited from only one parent, they can be used to trace the ancestry of individuals within a species without the complications caused by recombination between maternal and paternal genes.

Mitochondrial genes can also be used to trace the female ancestor of humans, while Y-chromosome genes can be used to trace male ancestors. In this way, differences between males and females in migration or patterns of reproduction can be detected. In addition, mitochondrial genes are used in studies of animal behavior to identify the parents of animals and birds and determine their social structure. Since the organelle genome is so highly simplified, mtDNA or cpDNA can be retrieved and analyzed from ancient or poorly preserved samples in which there would be no chance of retrieving a nuclear marker. SEE ALSO Cell, Eukaryotic; Mitochondrial Diseases; Mitochondrial Genome; Molecular Anthropology; Plasmid; Prion.

C. William Birky, Jr.

Bibliography

Cann, Rebecca L. "Genetic Clues to Dispersal in Human Populations: Retracing the Past from the Present." *Science* 291 (2001): 1742–1748.

Enserink, Martin. "Evolutionary Biology: Thanks to a Parasite, Asexual Reproduction Catches On." *Science* 275 (1997): 1743–1750.

Gray, Michael W., Gertraud Burger, and B. Franz Lang. "Mitochondrial Evolution." *Science* 283 (1999): 1476–1481.

Ochman, Howard, and Nancy A. Moran. "Genes Lost and Genes Found: Evolution of Bacterial Pathogenesis and Symbiosis." *Science* 292 (2001): 1096–1099.

Palmer, Jeffrey D. "Organelle Genomes: Going, Going, Gone!" *Science* 275 (1997): 790.

Wallace, Douglas C. "Mitochondrial Diseases in Man and Mouse." *Science* 283 (1999): 1482–1488.

Yaffe, Michael P. "The Machinery of Mitochondrial Inheritance and Behavior." *Science* 283 (1999): 1493–1497.

Inheritance Patterns

Inheritance patterns are the predictable patterns seen in the transmission of genes from one generation to the next, and their expression in the organism that possesses them. (A gene is said to be expressed when it is read by cellular mechanisms that result in the production of a protein.) While people have long noted that offspring resemble parents, the formal description of inheritance patterns began with Gregor Mendel, whose discoveries laid the foundation for the modern understanding of genetic inheritance.

Phenotype and Genotype

An organism's observable characteristics, such as height, hair texture, skin color, or ear shape, are known as the **phenotype** of that organism. The phenotype is determined partly by the environment and partly by the set of genes that the organism inherited from its parents. Adult height, for instance, is due partly to nutrition (an environmental influence), and partly to a set of genes governing things such as rates of bone growth, sensitivity to specific hormones, and the like. Phenotype includes not only large-scale characteristics such as height, but every expressed trait, including the types and amounts of all the proteins produced in each cell in the body.

phenotype observable characteristics of an organism

The set of genes an organism inherits is known as its **genotype**. Genes are carried on chromosomes in the cell nucleus. Animals and most other multicellular organisms possess two sets of chromosomes in each cell, one set inherited from the mother, and one from the father. Such an organism is said to be **diploid**. In humans, the maternal and paternal sets each include 23 chromosomes, so humans have 46 chromosomes in each cell. Analysis shows that the maternal and paternal chromosome sets are virtually identical, and they can be matched up to form 23 pairs. One pair, however, may not be a pair at all. These are the sex chromosomes, so called because they determine the sex of the organism. In humans, the female carries two identical sex chromosomes, called X chromosomes, while the male carries two dissimilar chromosomes, one X and one Y. The other 22 pairs of chromosomes are called **autosomes**.

genotype set of genes present

diploid possessing pairs of chromosomes, one member of each pair derived from each parent

autosomes chromosomes that are not sex determining (not X or Y)

Alleles

Members of each chromosome pair (except for X and Y) carry the same set of genes, so that a diploid cell carries two copies of (almost) every gene, one on the maternally derived chromosome, and one on the paternally derived chromosome. These two copies may be precisely identical, meaning the two genes have precisely the same sequence of nucleotides, or their sequences may be slightly different. These sequence differences may have no effect at all on the phenotype, or they may lead to different forms of the same trait, such as brown versus blue eye color, or smooth versus wrinkled pea texture. The two different forms of the gene are called **alleles**, and so we speak of the brown eye color allele or the wrinkled pea texture allele.

alleles particular forms of genes

While a single organism can possess no more than two different alleles for a single gene, many different alleles for a particular gene can exist in a population. For instance, there are three alleles for the ABO blood group gene, namely A, B, and O.

Dominance Relations

When both alleles for a particular trait are identical, the organism is said to be **homozygous** for that trait. When the alleles differ, the organism is **heterozygous**. The presence of two different alleles raises the question of whether one or the other, or both, will determine the phenotype of the organism. For his experiments on peas, Mendel chose traits for which one allele of each pair had a decisive effect, completely determining the phenotype even in the presence of the other allele. He called the determining allele "dominant" and the other allele "recessive." Only when the organism is homozygous for the recessive allele does the phenotype show the recessive trait. For instance, the albino skin pigmentation allele is recessive to other pigmentation alternatives.

homozygous containing two identical copies of a particular gene

heterozygous characterized by possession of two different forms (alleles) of a particular gene

While some sets of alleles do show a complete dominance-recessiveness relationship, most allele sets do not. Instead, each allele contributes to the phenotype. Such a relationship is called codominance. In humans, the A and B blood group alleles are codominant, and a person inheriting both will have blood type AB. Incomplete dominance is another variant in this system. In this case, the phenotype of the heterozygote is intermediate between the two extremes. Skin color in humans often shows this pattern.

Molecular Meaning of Dominance and Recessiveness

The terms "dominant" and "recessive" imply some competitive interaction between alleles over which one will control the phenotype. This is not the case, however. Alleles do not interact in the nucleus. Instead, both alleles are expressed (in most cases), and the phenotype reflects the result. How, then, can one allele determine the phenotype to the exclusion of another?

Genes code for proteins, and their effect on the phenotype is through the proteins they create. By analyzing the quantity and characteristics of protein produced from each allele, it has been shown that, in many cases, recessive alleles code for defective proteins, or for very low levels of protein. If the other allele codes for normal functional protein, and if the organism can make do with half the normal level of protein (or can increase production from the normal allele), the defective allele will have no effect on the phenotype, and will therefore act in a recessive manner. This type of allelic change is called a loss-of-function mutation. This is the case with the albino allele. This allele codes for a nonfunctional enzyme in the pathway that produces the pigment melanin. Even with only one functioning allele, the organism can still make enough melanin to obtain normal skin pigmentation. Thus, the functional allele will appear to be **dominant**.

dominant controlling the phenotype when one allele is present

Alleles coding for defective protein can act in a dominant manner in other situations, however. If the resulting protein is not properly regulated by other cell components, it may perform actions that harm the cell. This is called a toxic-gain-of-function mutation. Huntington's disease is thought to be due to a toxic-gain-of function mutation, although it is not yet clear what the exact toxic mechanism is.

A defective protein can also have a dominant effect if its absence cannot be compensated for by the other functioning allele (this is called a dominant negative effect). This may occur when half the normal protein level is insufficient for normal function (a condition called haploinsufficiency), or when the protein forms part of a multiprotein complex, which is therefore defective in its entirety. Examples include a variety of human collagen-structure disorders. Collagen is the most abundant and important structural protein in the body, and is critical for bone formation and growth. Defects in one of the subunits cause a variety of dominant disorders termed "osteogenesis imperfecta."

Autosomal Dominant Inheritance

Autosomal dominant inheritance is due to a dominant allele carried on one of the autosomes. Autosomal dominant alleles need only be inherited from one parent, either the mother or the father, in order to be expressed in the phenotype. Because of this, any child has a 50 percent chance of inheriting the allele and expressing the trait if one parent has it.

Many normal human traits are due to autosomal dominant alleles, including the presence of dimples, a cleft chin, and a widow's-peak hairline. Note that dominant does not necessarily mean common. Dominant alleles can be rare in a population, and do not spread simply because they are dominant. This phenomenon is explained by the theory known as Hardy-Weinberg equilibrium.

There are hundreds of medical conditions due to autosomal dominant alleles, most of them very rare. They include neurodegenerative disorders such as Huntington's disease, a variety of deafness syndromes, and metabolic disorders such as familial hypercholesterolemia (affecting blood cholesterol levels) and variegate porphyria (affecting the oxygen-carrying porphyrin molecule). Table 1 lists some other examples.

Because inheritance of a harmful dominant allele can be lethal, these alleles tend to be quite rare in the population, and new **mutations** account for many cases of these conditions. Exceptions include late-onset disorders such as Huntington's disease, in which parents may pass on the gene to offspring before developing the symptoms of the disease. Other exceptions arise from incomplete penetrance, in which the allele is present, but (for reasons usually unknown) it is not expressed. Genomic imprinting (see below) may explain some cases of incomplete penetrance. Variable expressivity is also possible, in which different individuals express the trait with different levels of severity.

mutations changes in DNA sequences

Condition	Chromosome Location and Inheritance Pattern	Protein Affected	Symptoms and Comments
Gaucher Disease	1, recessive	glycohydrolase glucocerebrosidase, a lipid metabolism enzyme	Common among European Jews. Lipid accumulation in liver, spleen, and bone marrow. Treat with enzyme replacement
Achondroplasia	4, dominant	fibroblast growth factor receptor 3	Causes dwarfism. Most cases are new mutations, not inherited
Huntington's Disease	4, dominant	huntingtin, function unknown	Expansion of a three-nucleotide portion of the gene causes late-onset neurodegeneration and death
Juvenile Onset Diabetes	6, 11, 7, others	IDDM1, IDDM2, GCK, other genes	Multiple susceptibility alleles are known for this form of diabetes, a disorder of blood sugar regulation. Treated with dietary control and insulin injection
Hemochromatosis	6, recessive	HFE protein, involved in iron absorption from the gut	Defect leads to excess iron accumulation, liver damage. Menstruation reduces iron in women. Bloodletting used as a treatment
Cystic Fibrosis	7, recessive	cystic fibrosis transmembrane regulator, an ion channel	Sticky secretions in the lungs impairs breathing, and in the pancreas impairs digestion. Enzyme supplements help digestive problems
Friedreich's Ataxia	9, recessive	frataxin, mitochondrial protein of unknown function	Loss of function of this protein in mitochondria causes progressive loss of coordination and heart disease
Albinism	11, recessive	tyorsinase	Lack of pigment in skin, hair, eyes; loss of visual acuity
Best Disease	11, dominat	VMD2 gene, protein function unknown	Gradual loss of visual acuity
Sickle Cell Disease	11, recessive	hemoglobin beta subunit, oxygen transport protein in blood cells	Change in hemoglobin shape alters cell shape, decreases oxygen-carrying ability, leads to joint pain, anemia, and infections. Carriers are resistant to malaria. About 8% of US black population are carriers
Phenylketonuria	12, recessive	phenylalanine hydroxylase, an amino acid metabolism enzyme	Inability to breakdown the amino acid phenylalanine causes mental retardation. Dietary avoidance can minimize effects. Postnatal screening is widely done
Marfan Syndrome	15, dominant	fibrillin, a structural protein of connective tissue	Scoliosis, nearsightedness, heart defects, and other symptoms
Tay-Sachs Disease	15, recessive	beta-hexosaminidase A, a lipid metabolism enzyme	Accumulation of the lipid GM2 ganglioside in neurons leads to death in childhood
Breast Cancer	17, 13	BRCA1, BRCA2 genes	Susceptibility alleles for breast cancer are thought to involve reduced ability to repair damaged DNA
Myotonic Dystrophy	19, dominant	dystrophia myotonica protein kinase, a regulatory protein in muscle	Muscle weakness, wasting, impaired intelligence, cataracts
familial hypercholesterolemia	19, incomplete dominance	low-density lipoprotein (LDL) receptor adenosine deaminase, nucleotide metabolism enzyme	Accumulation of cholesterol-carrying LDL in the bloodstream leads to heart disease and heart attack
Severe Combined Immune Deficiency ("Bubble Boy" Disease)	20, recessive	respiratory complex proteins	Immature white blood cells die from accumulation of metabolic products, leading to complete loss of the immune response. Gene therapy has been a limited success
Leber's Hereditary Optic Neuropathy	mitochondria, maternal inheritance	transfer RNA	degeneration of the central portion of the optic nerve, loss of central vision
Mitochondrial Encephalopathy, Lactic Acidosis, and Stroke (MELAS)	mitochondria, maternal inheritance	lignoceroyl-CoA ligase, in peroxisomes	recurring, stroke-like episodes in which sudden headaches are followed by vomiting and seizures; musle weakness
Adrenoleukodystrophy	X	dystrophin, muscle structural protein	Defect causes build-up of long-chain fatty acids. Degeneration of the adrenal gland, loss of myelin insulation in nerves. Featured in the film "Lorenzo's Oil"
Duchenne Muscular Dystrophy	X	Factor VIII, part of the blood clotting cascade	Lack of dystrophin leads to muscle breakdown, weakness, and impaired breathing
Hemophilia A	X		Uncontrolled bleeding, can be treated with injections or replacement protein
Rett Syndrome	X	methyl CpG-binding protein 2, regulates DNA transcription	Most boys die before birth. Girls develop mental retardation, mutism and movment disorder

Table 1.

Autosomal Recessive Inheritance

Autosomal recessive inheritance is due to recessive alleles carried on autosomes. An individual possessing only one recessive allele is known as a carrier. An individual must inherit two recessive alleles, one from each parent, in order to express the recessive trait. When two carrier parents have offspring, each offspring has a 25 percent chance of inheriting two alleles and expressing the trait. The two recessive alleles need not be precisely identical, as long as each is nonfunctional. An individual possessing two different alleles with the same effect is known as a compound heterozygote. Compound heterozygotes account for some cases of the neurologic disorder known as Friedreich's ataxia.

Medical conditions due to autosomal recessive traits also number in the many hundreds. These include cystic fibrosis (affecting ion transport in the lungs and pancreas), Tay-Sachs disease (affecting lipid metabolism and storage, especially in the brain), and hemochromatosis (affecting iron metabolism and storage in a variety of organs).

The number of people with such conditions is actually much higher than that for autosomal dominant conditions. This is because inheritance of one harmful recessive allele does not produce symptoms, and so the individual can reproduce and pass the allele on to children easily. Thus, most harmful recessive alleles are not deleted from a population's gene pool as rapidly as most dominant ones, and the likelihood of inheriting two copies is consequently higher. Most humans harbor a small handful of known harmful alleles; it is only when they mate with another who has the same set that there is a chance of bearing children that express the disorder. Customs that warn against marrying close relations have the effect of minimizing the likelihood of offspring with homozygous recessive conditions.

Sex-Linked Inheritance

The two sex chromosomes differ in the genes they carry. The Y chromosome is very small, and appears to carry very few genes other than the *SRY* gene that determines male sex. Many genes are carried on the X chromosome, however, and these are as essential for males as they are for females. Genes carried on the X chromosome are said to be X-linked.

X-linked dominant alleles affect both males and females, although males may be more severely affected since they inherit only a single X chromosome and thus lack a compensating normal allele. An example of a disorder caused by an X-linked dominant allele is congenital generalized hypertrichosis, which causes dense hair growth on the face and other regions in both sexes. X-linked dominant alleles can be inherited by both males and females, but fathers cannot pass them on to sons.

X-linked recessive alleles affect males more often and more severely than females. A male inherits his single X chromosome from his mother. Because a male has only one X chromosome, he expresses every allele on it, including harmful recessive ones. Examples of conditions due to recessive X-linked alleles include Duchenne muscular dystrophy, one form of hemophilia, and red-green colorblindness. These conditions are much more common in males than in females. Female carriers have a 50 percent chance of giving birth to a male child affected by the recessive allele. The genetic status of the father with respect to X-linked conditions is not relevant in this case, because he donates a Y chromosome to his male children.

Since females have two X chromosomes, they are less likely than males to express harmful recessive X-linked traits. Female children have only a 25 percent chance of inheriting two recessive alleles from a carrier mother and an affected father.

Mosaicism

The reason females are less often affected by recessive alleles is not as simple as it is for autosomes. Since females have twice the number of X chromosomes as males, the question arises as to whether they make twice the amount of each X-encoded protein as males do. In fact they do not, and are prevented from doing so by the random inactivation of one X chromosome in each cell. Therefore, about half of the heterozygote female's cells will express the normal allele, and half will express the harmful recessive allele. This is in contrast to the situation for autosomes, in which each cell expresses both alleles.

Inactivation begins early in development, with some cells shutting down one X and others shutting down the other, followed by faithful inheritance of the inactivated chromosome by each daughter cell following cell division. As a result, many tissues in the adult female are a mosaic of cells with different X chromosomes inactivated. The consequence of this is seen in a woman who is heterozygous for a harmful allele. If the affected tissue is primarily composed of cells expressing only the harmful one, she is likely to express the trait. However, most adult tissues will be a more even mixture, and for many disorders this will prevent her from developing symptoms of the disease.

The trait may also show variable expressivity, with the severity dependent on the proportion of the tissue affected. Duchenne muscular dystrophy in females is an example. Many women with one disease allele will show no symptoms. Others will develop only slightly elevated blood levels of certain enzymes indicating mild muscle damage, while others will develop heart problems and muscle weakness. Such women are referred to as "manifesting carriers."

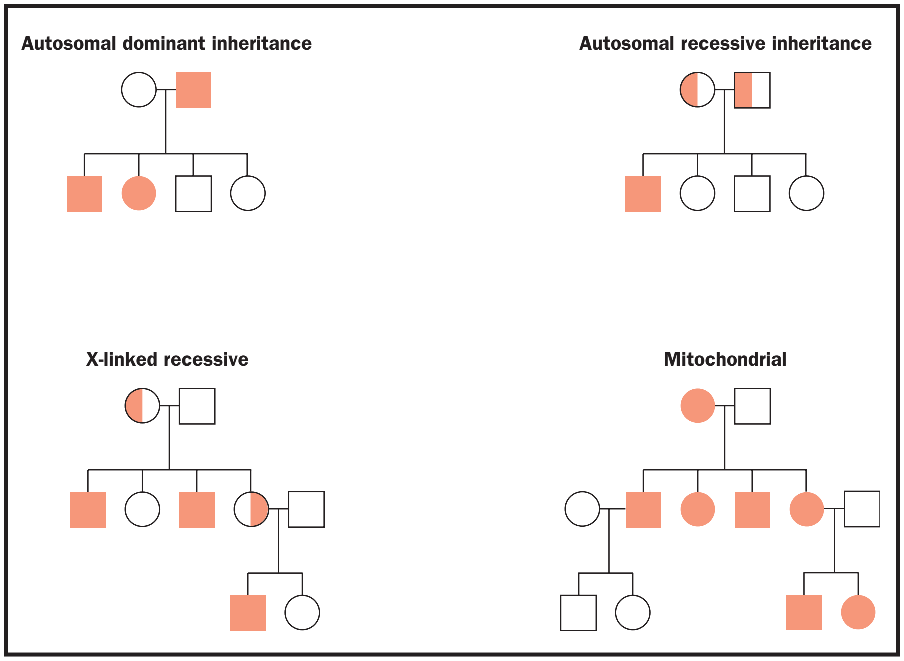
Mitochondrial Inheritance

Mitochondria are the cell's power plants. They possess their own chromosome, which carries thirty-seven genes. Mitochondria are inherited only from the mother. Mitochondrially inherited disorders include a number of rare muscle diseases (mitochondrial myopathies), as well as some deafness syndromes, optic nerve degeneration, and other neurological disorders.

Penetrance, Expressivity, and Anticipation

The presence of a dominant allele, or two recessive alleles, is not always a guarantee that the trait will be displayed in the phenotype, a phenomenon called incomplete penetrance. Variable expressivity also occurs, with some individuals more affected than others of the same genotype. In most cases the reasons for these differences are unknown, but they are assumed to be due at least in part to other differences between individuals. For instance, if there are differences between individuals in the other genes with which the product of incompletely penetrant or variably expressed allele interacts, this may account for some of these differences in expression.

One phenomenon also associated with changes in both timing and severity of expression is anticipation. Anticipation refers to the successive decrease in the age of symptom onset over several generations, so that a condition might first manifest at age 60 in a grandfather, at age 40 in a father, and at age 20 in a son. This increasingly earlier onset is often accompanied by



Mitochondrial inheritance passes only through the mother.

increasingly severe symptoms as well. Some cases of anticipation are known to be due to changes in the allele itself over time. Spinocerebellar ataxia, for example, is an autosomal dominant disease that causes balance disorders. The normal allele has a section of its DNA that includes approximately 20 repeats of a nucleotide triplet, CAG. The disease allele has 40 or more CAG repeats. This number can increase between generations, leading to earlier onset and more severe disease over several generations. Other so-called triplet repeat diseases show the same pattern of anticipation.

Imprinting

Some cases of incomplete penetrance appear to be due to imprinting. In this phenomenon, expression of an allele is governed by whether it is derived from the mother or the father. Imprinted alleles are located on autosomes, but are "stamped" with the sex of the parent that contributed it. The chemical basis of the imprint is the addition of methyl ($-CH_3$) groups to the allele's nucleotides, and the effect is thought to be to silence the allele so it is not expressed. For some genes the maternal allele is silenced, while for others the paternal allele is silenced. When a child inherits the two alleles, they retain these stamps, regardless of the sex of the child. However, when the child makes its own sperm or eggs, the child's own imprinting machinery stamps the alleles so that they correspond to the child's own sex. Therefore, a particular allele can be turned on or turned off as it is passed down through successive generations, from male to female and back again.

Imagine a dominant disease allele that is active when inherited from the mother, but is silenced when inherited from the father. Both the sons and daughters of the mother will develop symptoms of the disease. The daughter's children will also develop symptoms, while the son's children will not, despite having the same genotype. This is an example of incomplete penetrance. Prader-Willi syndrome and Angelman syndrome are examples of disorders arising from imprinted genes.

The pale skin, hair, and eyes typical of an albino presents in this young man, whose genetic makeup determined that he would not have the same dark skin pigment as his father.

Polygenic, Multifactorial, and Complex Traits

Proteins, which are the products of genes, interact with one another in complex ways to determine the phenotype. Almost every trait we observe, such as height, normal metabolic level, or intelligence, is really the product of many genes. Many traits, however, also reflect the influence of the environment. Such traits are called complex traits, to distinguish them from simple traits that are governed by single genes. While any single gene contributing to a complex trait can be described in terms of dominance or recessiveness, autosomal or sex-linked, or other categories, the gene products interact to make a much more subtle phenotypic picture.

A polygenic trait is a complex trait controlled by the alleles of two or more genes, without the influence of the environment. A multifactorial trait is a complex trait controlled by both genes and the environment. Intelligence is multifactorial, with strong influences from both genes (such as those controlling nerve-cell growth and connectivity) and the environment (such as early childhood nutrition and education).

As the number of influences grows, so too does the number of possible phenotypes. Because of this, complex traits show not just a few phenotypes, but a continuum (as can be seen in the wide range of possible human heights). The distribution in a population will usually be described by a bell-shaped curve, with most people displaying the mid-range phenotype.

Pleiotropy and Epistasis

Most single genes affect more than one observable trait, a phenomenon know as **pleiotropy**. For example, the alleles for melanin pigment affect skin color, eye color, and hair color. The ion channel gene affected in cystic fibrosis acts in the lungs, the pancreas, and other passageways, and defects cause symptoms in both these organs, as well as elsewhere in the body.

pleiotropy genetic phenomenon in which alteration of one gene leads to many phenotypic effects